

Enterprise SONiC Distribution by Dell Technologies Proof of Concept Deployment Guide

H18645.3

Abstract

This guide demonstrates a proof-of-concept (POC) deployment of a leaf-spine network with a BGP EVPN overlay and additional features such as multihoming, MC-LAG, VRF, and layer 2 and layer 3 concepts using Enterprise SONiC. The guide describes the topology requirements and processes for rapid deployment of a test network.

Dell Technologies Solutions

Notes, cautions, and warnings

 **NOTE:** A NOTE indicates important information that helps you make better use of your product.

 **CAUTION:** A CAUTION indicates either potential damage to hardware or loss of data and tells you how to avoid the problem.

 **WARNING:** A WARNING indicates a potential for property damage, personal injury, or death.

Chapter 1: Introduction.....	4
Introduction to Dell SONiC.....	4
Chapter 2: POC Requirements.....	5
Hardware requirements.....	5
Topology requirements.....	5
Prerequisites.....	6
POC reference deployment diagram.....	6
POC deployment features.....	9
Chapter 3: POC Deployment	11
POC deployment steps.....	11
Chapter 4: BGP EVPN VXLAN with MC-LAG Configuration.....	15
Spine configuration steps.....	15
BGP and BFD unnumbered configuration.....	16
East-West connectivity – Leaf1 and Leaf2 configuration.....	17
Virtual Routing Forwarding (VRF) configuration.....	19
MultiChassis Link Aggregation (MC-LAG) configuration.....	20
EVPN VXLAN overlay configuration.....	21
North-South connectivity - Border leaf configuration	23
Monitoring the POC environment.....	27
Verifying the POC.....	27
Chapter 5: BGP EVPN VXLAN with Multihoming Configuration	30
Spine configuration steps.....	30
BGP and BFD unnumbered configuration.....	31
East-West connectivity – Leaf switches configuration.....	32
VRF configuration.....	33
EVPN Multihoming (E-MH) configuration.....	34
North-South connectivity - Border leaf configuration.....	37
Monitoring the PoC environment.....	40
Verifying the PoC.....	40
Appendix A: Host Configuration.....	43
Ubuntu host configuration.....	43
Appendix B: POC End Host Configurations.....	44
Dell switch (Host1).....	44
Dell switch (Host2).....	45
Appendix C: Additional Information.....	46
Dell Technologies documentation.....	46
Support and feedback.....	46

Introduction

This chapter presents the following topics:

Topics:

- [Introduction to Dell SONiC](#)

Introduction to Dell SONiC

Enterprise SONiC Distribution by Dell Technologies is a hardened, validated, and supported version of the Software for Open Networking in the Cloud (SONiC) operating system for switch configuration and monitoring. It includes distribution of open-source community SONiC and additional features to support the ecosystem and partners.

Enterprise SONiC supports an industry-standard, intuitive Management Framework Command-Line Interface (MF-CLI) as well as programmatic interfaces such as REST API and the Google Remote Procedure Call (gRPC) Network Management Interface (gNMI). This guide uses MF-CLI to configure the network nodes.

The purpose of this guide is to demonstrate the procedures to configure a BGP leaf-spine topology with an EVPN overlay to support L2 multirack and multitenancy capabilities. Symmetric routing is used between networks in the same VRF. This topology meets many of the requirements for a modern data center to support segmentation and mobility of virtual machines (VMs).

POC Requirements

This chapter presents the following topics:

Topics:

- [Hardware requirements](#)
- [Topology requirements](#)
- [Prerequisites](#)
- [POC reference deployment diagram](#)
- [POC deployment features](#)

Hardware requirements

This POC uses the following Dell data center switches.

Dell PowerSwitch S5232F-ON

The PowerSwitch S5232F-ON switch is a 1-Rack Unit (1U), multilayer switch with 32 QSFP28 ports. The high-performance S5232F-ON switch is an optimal leaf or spine switch for environments requiring connectivity for 10/25/40/50/100 GbE ports.

In this guide, the PowerSwitch S5232-ON is used as a spine switch.



Figure 1. PowerSwitch S5232F-ON switch

Dell PowerSwitch S5248F-ON

The PowerSwitch S5248F-ON switch is a 1-Rack Unit (1U), multilayer switch with 48 SFP28 25 GbE ports, two QSFP28-DD ports (two 100 GbE interfaces per port), and four QSFP28 100 GbE ports. The high-performance S5248F-ON switch is an optimal leaf switch for environments that require connectivity for 25 GbE and 10 GbE compute and storage.

In this guide, the PowerSwitch S5248F-ON is used as a leaf switch.



Figure 2. PowerSwitch S5248F-ON switch

Topology requirements

The general requirements to set up the PoC topology include the following PowerSwitch switches with Open Network Install Environment (ONIE) installed, racked, and powered on:

- Two S5232F-ON as spines
- Four S5248F-ON as leaves

Prerequisites

- One S3248T-ON as a 1 GB management switch (or other suitable 1 GB switch)
- An optional DHCP/DNS server for automatic installation or manual installation with a static IP address
- A valid Dell Digital Locker (DDL) account with entitlement to Enterprise SONiC Distribution by Dell Technologies (see [Dell Support](#))
- An HTTP, TFTP, or FTP server connected to the management network
- Two servers to be used as fabric host devices with two 10 GbE or 25 GbE network interfaces
- Supported cables and optics (see [Support Matrix for Enterprise SONiC Distribution by Dell Technologies 3.2.](#))
- Console connection to a terminal server or laptop: Required serial port settings are 115200, 8 data bits, and no parity

The following figure shows the ideal physical topology for the management network:

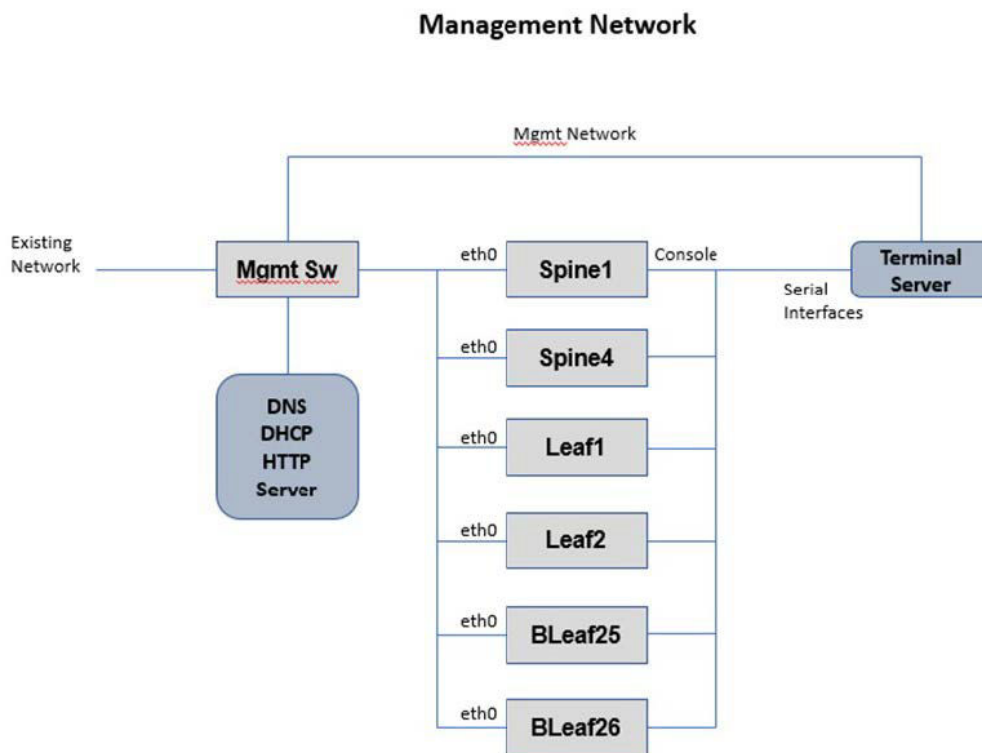


Figure 3. Management network topology

NOTE:

This topology reflects the interface and node naming convention used in the configuration.

POC reference deployment diagram

This POC uses a typical leaf and spine topology. The leaf switches are PowerSwitch S5248F-ON switches, while the spine switches are PowerSwitch S5232F-ON switches.

MultiChassis Link Aggregation (MC-LAG) is configured on the leaf switches to provide link and device redundancy at the leaf layer.

The following figure shows the physical connections that are used in the configurations for this POC.

NOTE: The external router is shown only for reference and is not used in this deployment.

NOTE:

Host 2 connected to border leaf switches is ONLY for a POC setup. In a production environment, hosts or workloads do not connect to border leaf switches.

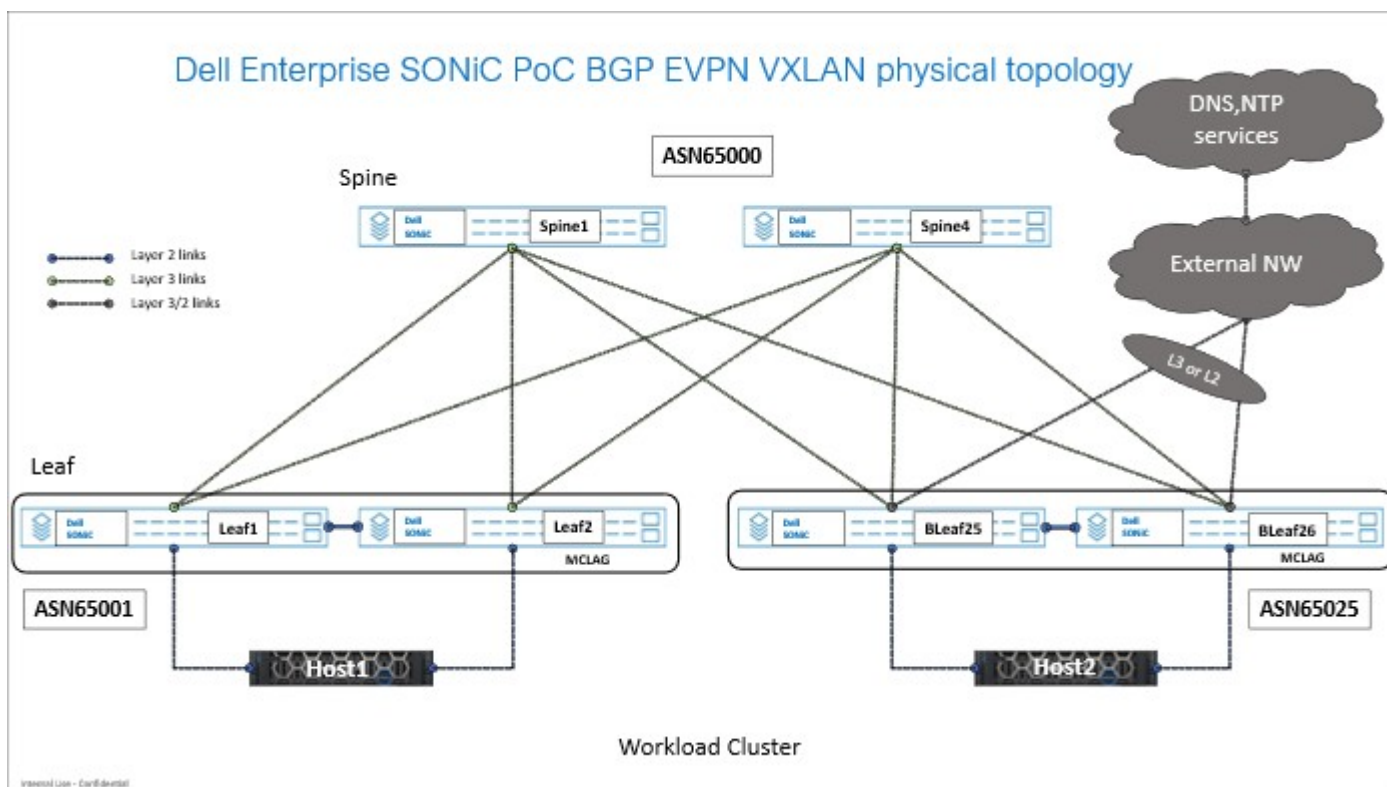


Figure 4. POC physical reference deployment setup with MC-LAG

The following figure shows the physical BGP EVPN VXLAN with EVPN multihoming topology PoC:

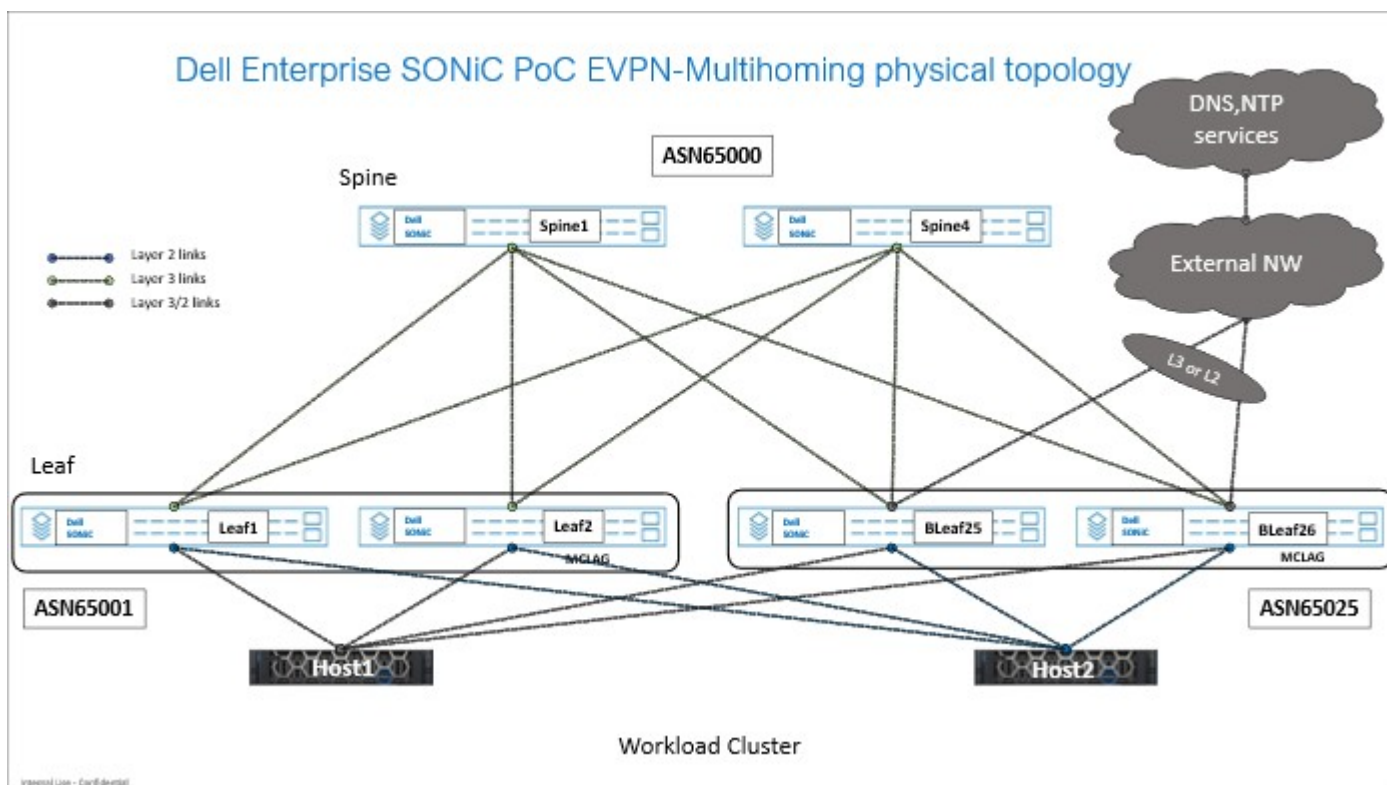


Figure 5. PoC physical reference deployment setup with EVPN multihoming

NOTE: This topology reflects the interface and node naming convention used in the configuration.

The following figure shows the logical layout of the underlay and overlay networks. Two Virtual Network Instances (VNIs) are deployed between the two leaf pairs depicting the overlay network. The L2 VNI supports stretching the L2 network (Vlan 140) between the leaf pairs. The L3 VNI supports symmetric routing between networks in the same VRF.

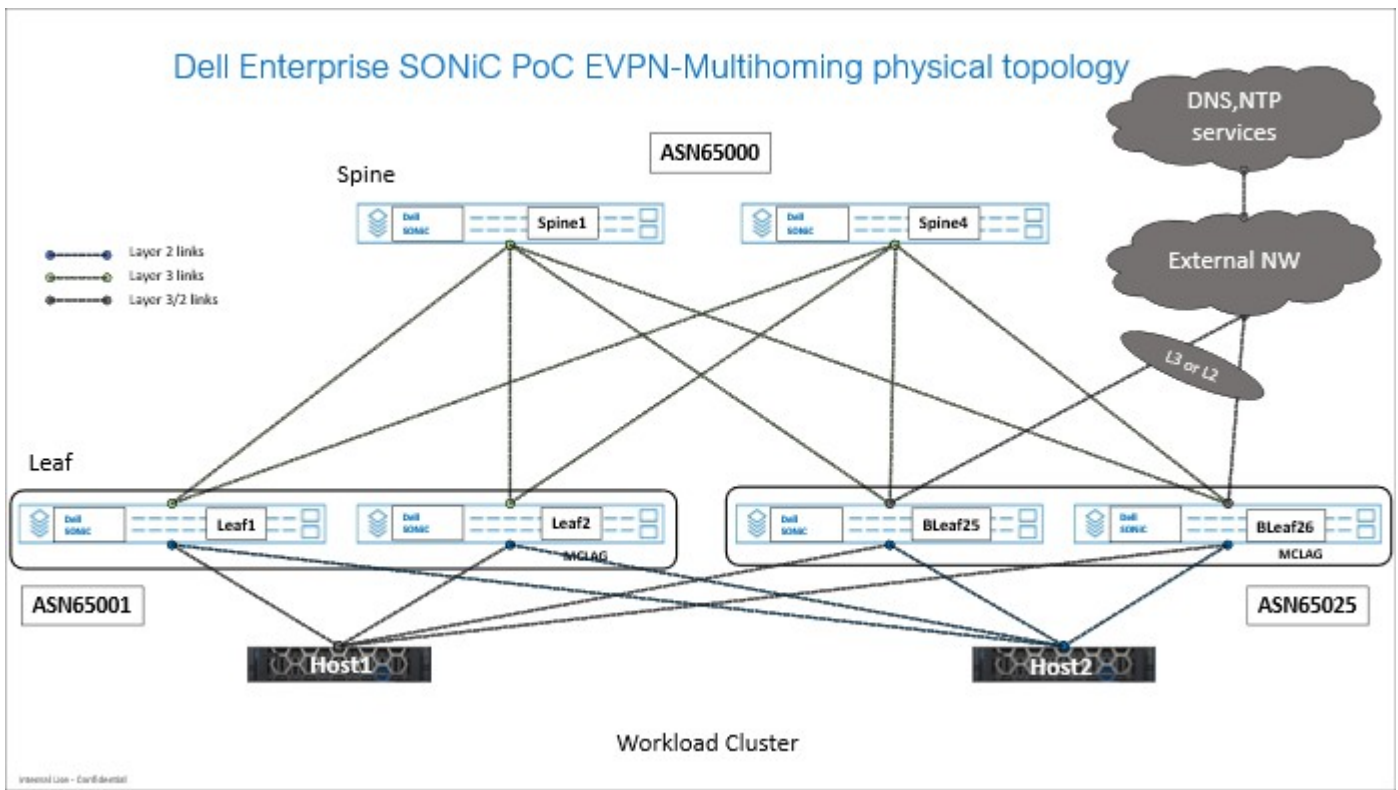


Figure 6. POC logical reference deployment setup

The following figure shows the overlay networks. Two VNIs are deployed between the two leaf pairs depicting the overlay network. The L2 VNI supports stretching the L2 network (Vlan 140) between the leaf pairs. The L3 VNI supports symmetric routing between networks in the same VRF.

In this topology, EVPN multihoming is deployed at the leaf access layer that is connected to the nodes. Notice the different connection methodologies from the nodes to the leaf switches as compared to an MC-LAG deployment (as shown in the preceding figure).

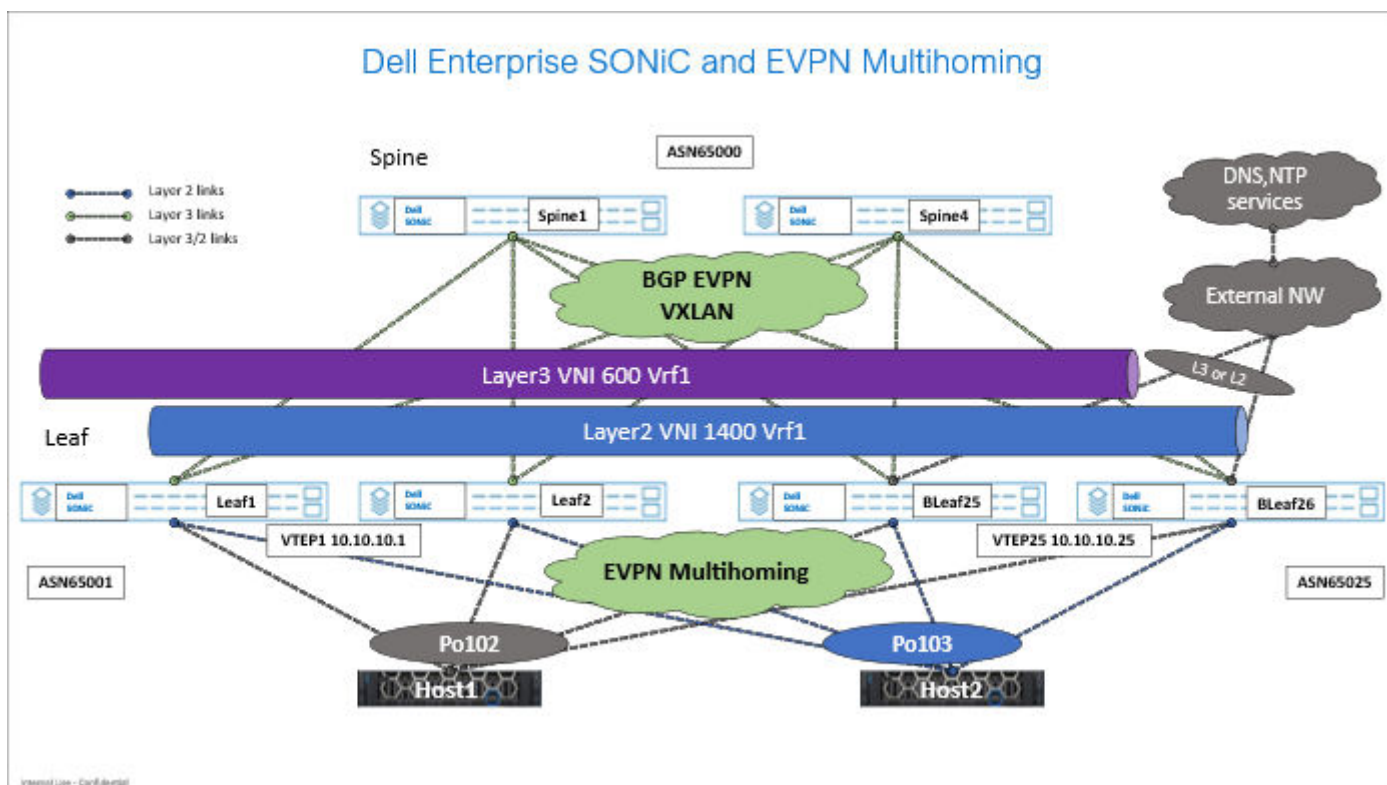


Figure 7. PoC logical reference deployment setup

In the preceding figures:

- The leaf and border leaf devices are configured to act as Virtual Tunnel End Points (VTEPs). One Layer 3 VNI is created per Virtual Routing and Forwarding (VRF). IPv4 prefixes learned on the VRF are redistributed into the overlay. The leaf devices are Layer 2 with the hosts.
- Border leaf 25 (BLeaf25) and border leaf 26 (BLeaf26) are peered to the gateway router that is then used to advertise the default route.
- Each host has two paths that connect to two leaf switches through a port channel.
- Each leaf has two paths that connect to the spine.
- Each leaf has a backup path that connects to the peer leaf.

POC deployment features

The following features are configured for this POC:

- BGP EVPN VXLAN
- MC-LAG
- EVPN Multihoming
- The interfaces that are used in this POC are 100 GbE except the interfaces to the servers, which are 10/25 GbE.
- LLDP is enabled by default.
- Layer 3:
 - All Layer 3 interfaces and VLANs with IP addresses are configured with an MTU of 9100 bytes.
 - The fabric is configured with BGP unnumbered.
 - eBGP is configured in the fabric.
 - ECMP is configured between the leaf and spine switches.
 - ASN allocations are unique on the leaf and spine switches:
 - Leaf 1 and Leaf 2—ASN# 65001
 - BLeaf 25 and BLeaf 26—ASN# 65025
 - Spine 1 and Spine 4—ASN# 65000
- VRF:

- VRF in leaf, single tenant setup:
 - Vrf-tenant1 (Tenant1) for overlay (EVPN) routes
 - Default VRF for underlay routes and connectivity
- Monitoring:
 - Syslog—Interface logs, BGP session logs, and chassis-related (PSU, fan, temperature) syslog alerts
 - SNMP—Enabled SNMPv2 queries and destination for SNMP traps
- IP address prefixes—The following table shows the node addressing information:

Table 1. IP Prefixes – Switches and VXLAN VTEP tunnels

Node	BGP ASN	Router-ID Loopback0	VTEP IP Loopback1	Stretched L2 Anycast-address	L3 VLAN Anycast-address	Anycast-MAC-address
Leaf1	65001	10.0.2.1	10.10.10.1	192.168.140.254/24	192.168.50.254/24	00:00:00:00:00:01:02
Leaf2	65001	10.0.2.2	10.10.10.1	192.168.140.254/24	192.168.50.254/24	00:00:00:00:00:01:02
BLeaf25	65025	10.0.2.25	10.10.10.25	192.168.140.254/24	192.168.80.254/24	00:00:00:00:00:01:02
BLeaf26	65025	10.0.2.26	10.10.10.25	192.168.140.254/24	192.168.80.254/24	00:00:00:00:00:01:02
Spine1	65000	10.0.1.1	N/A	N/A	N/A	N/A
Spine4	65000	10.0.1.4	N/A	N/A	N/A	N/A

POC Deployment

This chapter presents the following topics:

Topics:

- [POC deployment steps](#)

POC deployment steps

The following steps provide the framework for the POC deployment:

1. Download the Enterprise SONiC Distribution code:
 - a. Go to [Dell Software Support](#) and log in with your Dell registered credentials (DDL).
 - b. Select the appropriate entry:

Actions:	Select an action	GO	1-10 of 66	< 1 2 3 ... 7 >	10 per page
☐	Product name	Primary ID	License date	Order number / Dell Purchase Id	Details
☐	Enterprise SONiC Distribution, Premium, 1-year subscription, 1G platforms	Entitlement Id: DE00000285184961	05/27/2020	2020052705	Important Note
☐	Enterprise SONiC Distribution, Standard, 1-year subscription, 1G platforms	Entitlement Id: DE00000285184936	05/27/2020	2020052701	Important Note
☐	Enterprise SONiC Distribution, Standard, 1-year subscription, 400G platforms	Entitlement Id: DE00000285184953	05/27/2020	2020052708	Important Note

Figure 8. Enterprise SONiC entitlement

- c. Select the appropriate Enterprise SONiC entitlement for your switches and download the installer:

☒	Enterprise-SONIC-OS-3.0.1-Enterprise-Standard.zip	06/11/2020	3.0.1	Download
☐	Enterprise-SONIC-OS-3.0.1-Cloud-Standard.zip	06/11/2020	3.0.1	Download
☐	Enterprise-SONIC-OS-3.0.1-datamodel.zip	06/11/2020	3.0.1	Download

Figure 9. Enterprise SONiC images

To run this POC, select **Enterprise_SONiC_OS_3.2.0_Enterprise_Standard** version or later.

NOTE: The **Enterprise_Premium** version is also supported for this POC.

- d. Extract the .zip file and copy the .bin file to your HTTP, TFTP, or FTP server.
2. Uninstall any existing images.

When you reboot the switches, a startup menu is displayed. Use the arrow keys on the keyboard to select the ONIE :
Uninstall OS option:

```

+-----+
|*ONIE: Install OS
| ONIE: Rescue
| ONIE: Uninstall OS
| ONIE: Update ONIE
| ONIE: Embed ONIE
+-----+

```

Figure 10. ONIE startup switch menu: Uninstall OS

3. Install the newly downloaded Enterprise SONiC image:

- a. Use the arrow keys on the keyboard to select the ONIE: Install OS option:

```

+-----+
|*ONIE: Install OS
| ONIE: Rescue
| ONIE: Uninstall OS
| ONIE: Update ONIE
| ONIE: Embed ONIE
+-----+

```

Figure 11. ONIE startup switch menu: Install OS

The switch reboots.

- b. Run the `onie-discovery-stop` command to stop the ONIE discovery process:

```
ONIE:/ # onie-discovery-stop
```

- c. If a DHCP server is not available, manually configure an IP address (for example, 192.168.0.2). To configure an IP address on the switch management interface, run the following commands at the ONIE# prompt:

```
ONIE:/ # ifconfig eth0 192.168.0.2 netmask 255.255.255.0
ONIE:/ # ip route add default via 192.168.0.1
```

- d. Ensure that the switches can contact the HTTP server hosting the image before attempting the installation. Pinging the HTTP server verifies this.

```
ONIE:/ # ping 192.168.2.210
```

- e. Run the following command to install the Enterprise SONiC 3.2 code on the switches:

```
ONIE:/ # onie-nos-install https://192.168.2.210/
Enterprise_SONiC_OS_3.2.0_Enterprise_Standard.bin
```

When the image has been installed, the switches reboot with the correct image.

4. Log in to the switches:

- a. Use username **admin** and password **YourPaSsWoRd**:

```

At Console:
Debian GNU/Linux 9 sonic ttyS1

sonic login: admin
Password: YourPaSsWoRd

SSH from any remote server to sonic can be done by connecting to the IP address of the
Management interface
user@debug:~$ ssh admin@sonic_ip_address(or SONIC DNS Name)
admin@sonic's password:

```

Figure 12. Logging in to the switch

- b. Change the default password, as shown in the following figure:

```

You are required to change your password immediately (root enforced)
Changing password for admin.
(current) UNIX password: YourPaSsWoRd
Enter new UNIX password: *****
Retype new UNIX password: *****
Linux sonic 4.9.0-11-2-amd64 #1 SMP Debian 4.9.189-3+deb9u2 (2019-11-11) x86_64
You are on

/ _ _ | / _ _ | \ | } / _ _ |
\ _ _ | | _ _ | \ | } | _ _ |
| _ _ | \ _ _ | \ | } | _ _ |

-- Software for Open Networking in the Cloud --

Unauthorized access and/or use are prohibited.
All access and/or use are subject to monitoring.

Help:      http://azure.github.io/SONiC/

admin@sonic:~$

```

Figure 13. Changing the password from the default

5. Disable the Zero-Touch Provisioning (ZTP) process:

- a. Run the following command at the switch prompt:

```
admin@sonic:~$ sudo config ztp disable -y
```

The switches run ZTP when they first start up. This process must be disabled when the ZTP server is not being used to configure the switches. Allow a few minutes for the switch restart and allow manual configuration.

- b. Verify that the switches are ready by running the following command at the switch prompt:

```
admin@sonic:~$ show system status
System is ready
```

6. Perform the actions in this step only if Enterprise SONiC has not been recently loaded from ONIE:

- a. Ensure that the switches have set the default port profile.
- The correct HwSku for a PowerSwitch S5248F-ON is: DellEMC-S5248f-P-25G-DPB (Present).
 - The correct HwSku for a PowerSwitch S5232F-ON is: DellEMC-S5232f-C32 (Present).
- b. To view the effective port profile, run the HwSku command:

```
admin@sonic:~$ sudo config-hwsku list
```

- c. To change the port profile, run the appropriate command:

- For the PowerSwitch S5248F-ON, run:

```
admin@sonic:~$ sudo config-hwsku set DellEMC-S5248f-P-25G-DPB
```

- For the PowerSwitch S5232F-ON, run:

```
admin@sonic:~$ sudo config-hwsku set DellEMC-S5232f-C32
```

The switch configuration is cleared, and a reboot occurs.

7. Verify Layer 2 or Layer 3 mode on the ASIC.

NOTE: After you install Enterprise SONiC from ONIE, the default switching profile for a switch running Enterprise SONiC is Layer 3.

- a. If Enterprise SONiC was not installed from ONIE, verify the switching profile configuration:

```
admin@sonic:~$ sonic-cli
sonic# show running-configuration | grep factory
factory default profile l3 confirm
```

- b. If the switching profile is set to Layer 2, change the switching profile:

```
admin@sonic:~$ sonic-cli
sonic# configure terminal
sonic(config)# factory default profile l3 confirm
```

The switch reboots after the appropriate command is run.

8. If necessary, configure breakout ports:

i **NOTE:** This guide uses the default port breakouts. If any of the high-speed ports (Eth1/49 – Eth1/56 on a PowerSwitch S5248F-ON or Eth1/1 – Eth1/32 on a PowerSwitch S5232F-ON) must be broken out (4 x 25 GbE, 4x 10 GbE), do so before configuring the switch.

- a. Run the following commands:

```
admin@sonic:~$ sonic-cli
sonic# configure terminal
sonic(config)# interface breakout port 1/53 mode 4x25G
```

For the Dynamic Port Breakout in-progress, use `show interface breakout port 1/53` to check status message displays.

- b. Verify the success of the breakout by running:

```
sonic# show interface breakout port 1/53
-----
Port   Breakout Mode   Status           Interfaces
-----
1/53   4x25G           Completed        Eth1/53/1
                                   Eth1/53/2
                                   Eth1/53/3
                                   Eth1/53/4
```

9. If using 10 GbE server interfaces on a S5200 series switch, change the port group to support 10 GbE interfaces. These are applied to groups of four—for example, Eth1/1-1/4.

i **NOTE:** Only the S5212F-ON, S5224F-ON, S5248F-ON, and S5296F-ON switches have port groups.

For example, change interfaces Eth1/1 to Eth 1/4 from 25 GbE to 10 GbE:

```
admin@sonic:~$ sonic-cli
sonic# configure terminal
sonic(config)# port-group 1 speed 10000
sonic(config)# end
sonic# write memory
```

BGP EVPN VXLAN with MC-LAG Configuration

This chapter presents the following topics:

Topics:

- Spine configuration steps
- BGP and BFD unnumbered configuration
- East-West connectivity – Leaf1 and Leaf2 configuration
- Virtual Routing Forwarding (VRF) configuration
- MultiChassis Link Aggregation (MC-LAG) configuration
- EVPN VXLAN overlay configuration
- North-South connectivity - Border leaf configuration
- Monitoring the POC environment
- Verifying the POC

Spine configuration steps

The following diagram highlights the devices that are configured in this chapter. Spine1 and Spine4 are configured demonstrating the use of four spines.

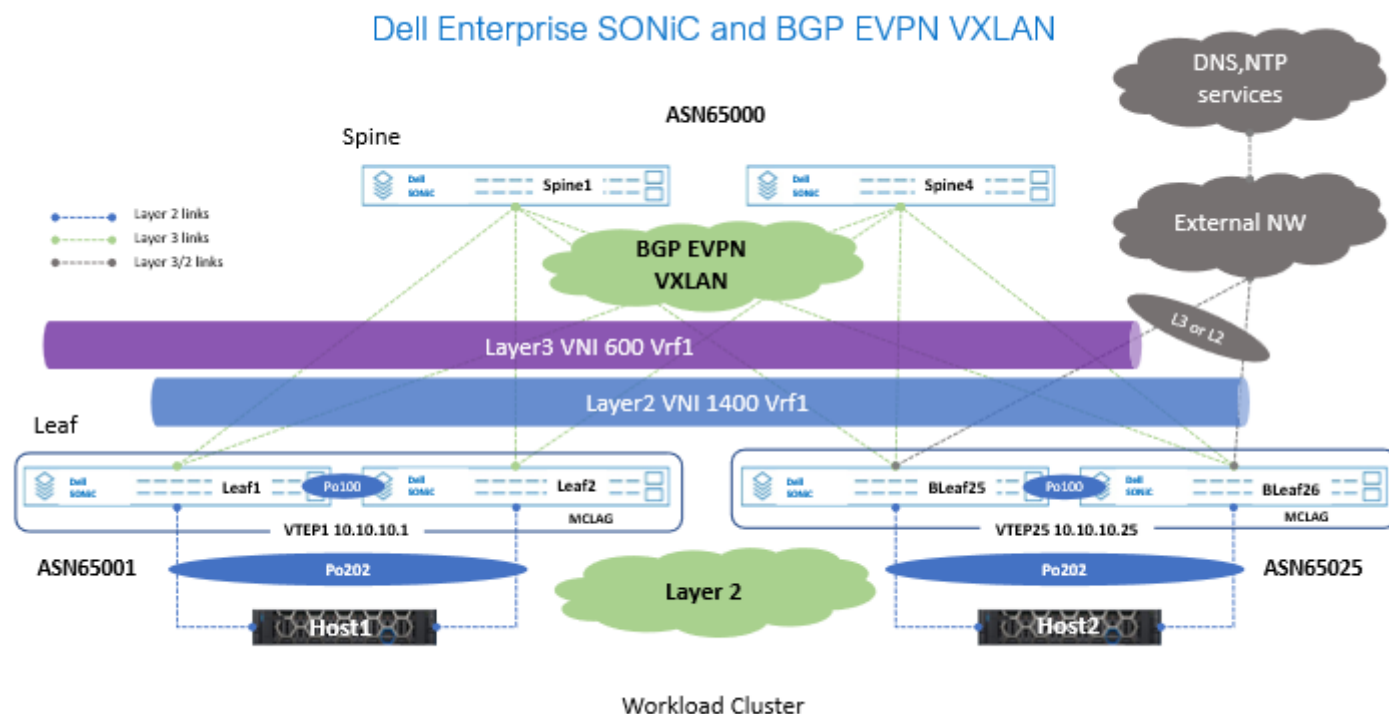


Figure 14. BGP EVPN VXLAN reference diagram

NOTE: Only Spine1 and Spine 4 are configured in this guide. More spines can be added if required to increase fabric bandwidth.

1. Enter Management Framework CLI (MF-CLI) and configuration mode.
2. Change interface naming mode to **Standard** and add hostname.
3. Exit back to the Linux shell to activate the changes.

Spine1	Spine4
<pre>sonic-cli configure terminal interface-naming standard</pre>	<pre>sonic-cli configure terminal interface-naming standard</pre>
<pre>hostname Spine1 end exit</pre>	<pre>hostname Spine4 end exit</pre>

4. Re-enter the MF-CLI and configuration mode.
5. Assign a loopback interface with a unique router ID for each spine.
6. Enable IPv6 on the spine downlinks to support unnumbered BGP point-to-point links and enable the interfaces.

Spine1	Spine4
<pre>sonic-cli configure terminal interface loopback 0 description router-id ip address 10.0.1.1/32 exit interface Eth 1/1 description Leaf1 ipv6 enable no shutdown interface Eth 1/2 description Leaf2 ipv6 enable no shutdown interface Eth 1/25 description BLeaf25 ipv6 enable no shutdown interface Eth 1/26 description BLeaf26 ipv6 enable no shutdown exit</pre>	<pre>sonic-cli configure terminal interface loopback 0 description router-id ip address 10.0.1.4/32 exit interface Eth 1/1 description Leaf1 ipv6 enable no shutdown interface Eth 1/2 description Leaf2 ipv6 enable no shutdown interface Eth 1/25 description BLeaf25 ipv6 enable no shutdown interface Eth 1/26 description BLeaf26 ipv6 enable no shutdown exit</pre>

BGP and BFD unnumbered configuration

1. Configure the BGP router with ECMP enabled.
2. Add the IPv4 unicast address-family to advertise the router-ID and paths for two leafs.
3. Configure the BGP leaf peer-group. Set timers and enable unnumbered BGP and BFD.
4. Activate both IPv4 unicast and EVPN address families for the peer group.
5. Assign the leaf neighbor interfaces to the peer-group.
6. Save the configuration.

Spine1	Spine4
<pre> router bgp 65000 router-id 10.0.1.1 bestpath as-path multipath-relax address-family ipv4 unicast redistribute connected maximum-paths 2 exit peer-group LEAF advertisement-interval 5 timers 3 9 remote-as external capability extended-nexthop bfd address-family ipv4 unicast </pre>	<pre> router bgp 65000 router-id 10.0.1.4 bestpath as-path multipath-relax address-family ipv4 unicast redistribute connected maximum-paths 2 exit peer-group LEAF advertisement-interval 5 timers 3 9 remote-as external capability extended-nexthop bfd address-family ipv4 unicast </pre>
<pre> activate exit address- family l2vpn evpn activate exit exit neighbor interface Eth 1/1 peer-group LEAF exit neighbor interface Eth 1/2 peer-group LEAF exit neighbor interface Eth 1/25 peer-group LEAF exit neighbor interface Eth 1/26 peer-group LEAF exit exit end write memory </pre>	<pre> activate exit address-family l2vpn evpn activate exit exit neighbor interface Eth 1/1 peer-group LEAF exit neighbor interface Eth 1/2 peer-group LEAF exit neighbor interface Eth 1/25 peer-group LEAF exit neighbor interface Eth 1/26 peer-group LEAF exit exit end write memory </pre>

East-West connectivity – Leaf1 and Leaf2 configuration

The following diagram highlights the leaf switches configured in this chapter. Leaf1 and Leaf2 are configured as an MC-LAG pair.

Dell Enterprise SONiC and BGP EVPN VXLAN

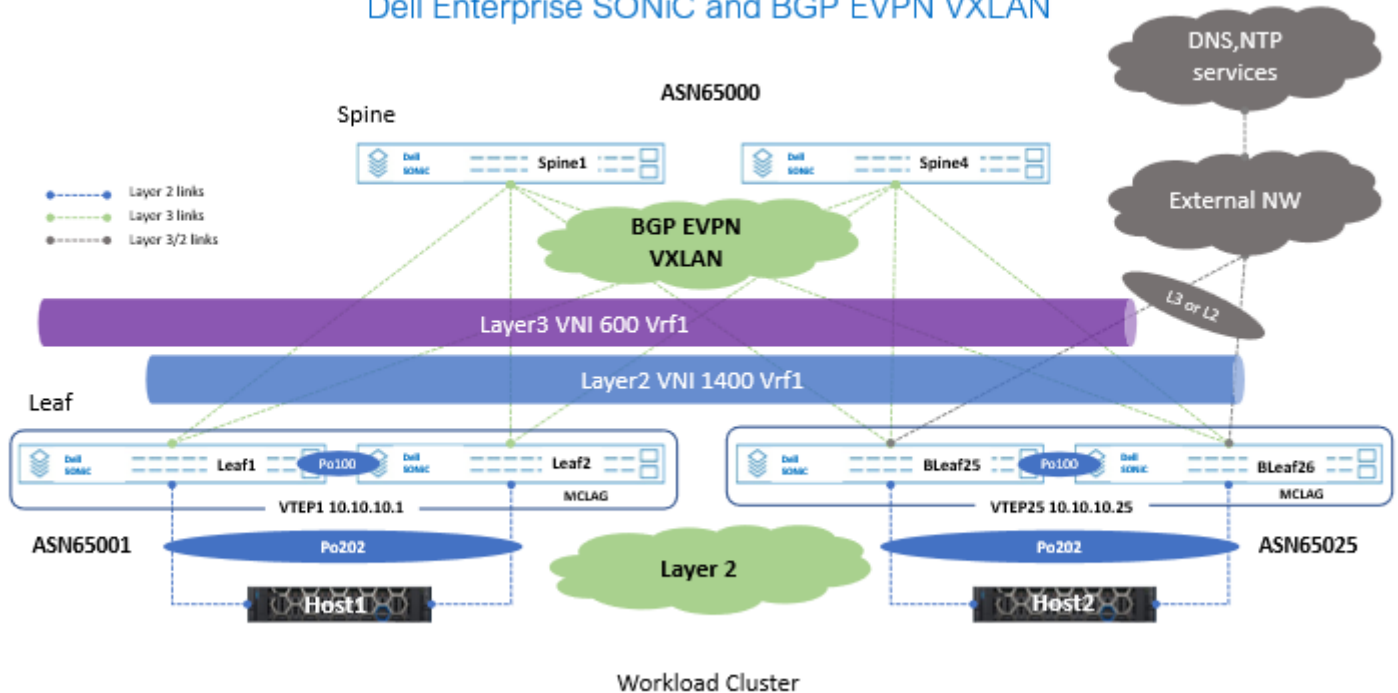


Figure 15. Leaf1 and Leaf2 configuration

1. Enter the MF-CLI and then configuration mode.
2. Change the interface naming mode to **Standard** and add a hostname.
3. Exit back to Linux shell to activate the changes.

Leaf1	Leaf2
<pre>sonic-cli configure terminal interface-naming standard hostname Leaf1 end exit</pre>	<pre>sonic-cli configure terminal interface-naming standard hostname Leaf2 end exit</pre>

4. Re-enter the MF-CLI and configuration mode.
5. Assign a loopback interface with a unique router ID on each leaf.
6. Then enable IPv6 on spine uplinks to support unnumbered BGP point-to-point links and enable the interfaces.

Leaf1	Leaf2
<pre>sonic-cli configure terminal interface loopback 0 description router-id ip address 10.0.2.1/32 exit interface Eth 1/53 description Spine1 ipv6 enable no shutdown interface Eth 1/56</pre>	<pre>sonic-cli configure terminal interface loopback 0 description router-id ip address 10.0.2.2/32 exit interface Eth 1/53 description Spine1 ipv6 enable no shutdown</pre>

Leaf1	Leaf2
<pre> description Spine4 ipv6 enable no shutdown exit </pre>	<pre> interface Eth 1/56 description Spine4 ipv6 enable no shutdown exit </pre>

Virtual Routing Forwarding (VRF) configuration

1. Assign a VRF for each tenant. This POC has only one tenant, but will support multiple tenants with additional VRFs.
2. Configure VLANs 50 and 140 for the hosts, bind to the tenant's VRF, and assign the same anycast-address to each host VLAN.
3. Assign a dedicated VLAN (Vlan60) for a L3 VNI and bind to the tenant's VRF.

NOTE: The VLAN assigned to the L3 VNI does not need an IP address.

Leaf1	Leaf2
<pre> ip vrf Vrf-tenant1 interface Vlan 50 ip vrf forwarding Vrf-tenant1 ip anycast-address 192.168.50.254/24 neigh-suppress exit interface Vlan 140 ip vrf forwarding Vrf-tenant1 ip anycast-address 192.168.140.254/24 neigh-suppress exit interface Vlan 60 ip vrf forwarding Vrf-tenant1 exit </pre>	<pre> ip vrf Vrf-tenant1 interface Vlan 50 ip vrf forwarding Vrf-tenant1 ip anycast-address 192.168.50.254/24 neigh-suppress exit interface Vlan 140 ip vrf forwarding Vrf-tenant1 ip anycast-address 192.168.140.254/24 neigh-suppress exit interface Vlan 60 ip vrf forwarding Vrf-tenant1 exit </pre>

4. Configure the MC-LAG peer-link and port channel members.
5. Configure all VLANs on the MC-LAG's peer-link port channel.

NOTE: The MC-LAG peer-link must include the VLAN of the L3 VNIs.

Leaf1	Leaf2
<pre> interface PortChannel 100 description MCLAG-Peer-Link switchport trunk allowed vlan add 50,60,140 exit interface Eth 1/49 description MCLAG-Peer-Link channel-group 100 no shutdown exit interface Eth 1/51 description MCLAG-Peer-Link channel-group 100 no shutdown exit </pre>	<pre> interface PortChannel 100 description MCLAG-Peer-Link switchport trunk allowed vlan add 50,60,140 exit interface Eth 1/49 description MCLAG-Peer-Link channel-group 100 no shutdown exit interface Eth 1/51 description MCLAG-Peer-Link channel-group 100 </pre>

Leaf1	Leaf2
	<u>no shutdown</u> <u>exit</u>

CAUTION: When copying and pasting the content from the preceding table, ensure that the underlined monospace text remains on one line. Failure to keep the underlined content on one line results in a line error on the switch.

MultiChassis Link Aggregation (MC-LAG) configuration

Configure MC-LAG based on the following diagram:

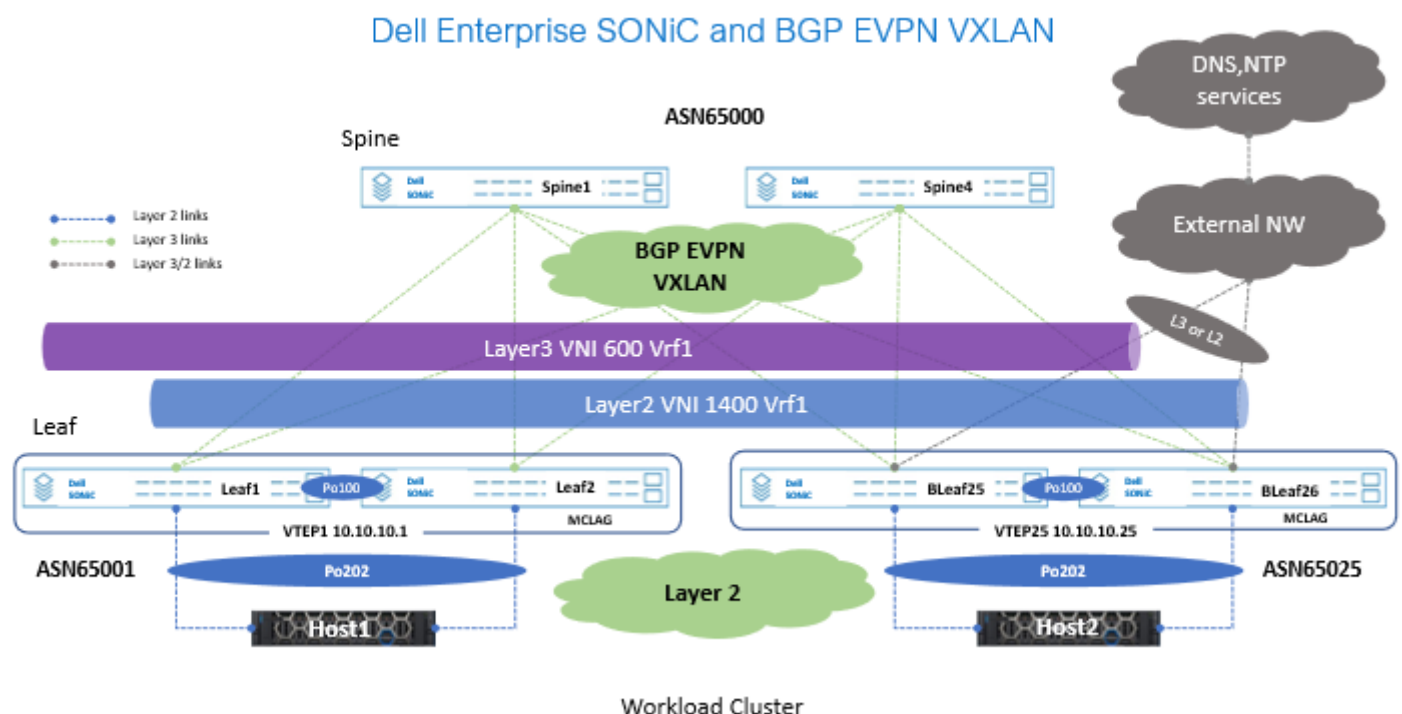


Figure 16. MC-LAG configuration

1. Configure the MC-LAG domain.
2. The loopback IP address that is used for the router-id is also used as the MC-LAG source and peer IP address.
3. Assign the host's port-channel interfaces and assign to the MC-LAG domain.

NOTE: By default, port channels are LACP. Use `interface PortChannel <num> mode on` for static LAGs. If changing LAG modes, you must completely remove and reassign the port channel.

NOTE: The spine switches are used for the MC-LAG peer keep-alive link communication

Leaf1	Leaf2
<pre> mclag domain 1 source-ip 10.0.2.1 peer-ip 10.0.2.2 peer-link PortChannel 100 delay-restore 90 exit interface PortChannel 202 description Host1_PortChannel switchport trunk allowed vlan add 50,140 mclag 1 exit interface Eth 1/3 description Host1_PortChannel channel-group 202 no shutdown exit </pre>	<pre> mclag domain 1 source-ip 10.0.2.2 peer-ip 10.0.2.1 peer-link PortChannel 100 delay-restore 90 exit interface PortChannel 202 description Host1_PortChannel switchport trunk allowed vlan add 50,140 mclag 1 exit interface Eth 1/3 description Host1_PortChannel channel-group 202 no shutdown exit </pre>

EVPN VXLAN overlay configuration

1. Enable static anycast-address and assign a unique MAC to be used across all leaf switches.
2. Create a loopback interface for the VTEP IP address. This address must be the same for each MC-LAG peer.
3. Create a VXLAN interface and assign the same VTEP source IP address to each peer.
4. Assign a unique primary-ip to optimize routing for orphan ports and active-standby hosts. The router-id can be used for this purpose.
5. Map the L2 VNI to the host VLANs that are stretched across the fabric.
6. Create an L3 VNI by mapping the dedicated VLAN (Vlan60) to the non-default VRF.

NOTE: The Network Virtualization Overlay (NVO) is automatically assigned to the VXLAN interface.

Leaf1	Leaf2
<pre> ip anycast-address enable ip anycast-mac-address 00:00:00:00:01:02 interface Loopback 1 description LogicalVTEP ip address 10.10.10.1/32 exit interface vxlan vtep1 source-ip 10.10.10.1 primary-ip 10.0.2.1 map vni 1400 vlan 140 map vni 600 vlan 60 map vni 600 vrf Vrf-tenant1 exit </pre>	<pre> ip anycast-address enable ip anycast-mac-address 00:00:00:00:01:02 interface Loopback 1 description LogicalVTEP ip address 10.10.10.1/32 exit interface vxlan vtep1 source-ip 10.10.10.1 primary-ip 10.0.2.2 map vni 1400 vlan 140 map vni 600 vlan 60 map vni 600 vrf Vrf-tenant1 exit </pre>

7. Configure the BGP router with ECMP enabled.
8. Configure the underlay's address-family and redistribute the connected networks.
9. Add VNI advertisements to the EVPN address family and assign the MC-LAG peer's primary-ip.

Leaf1	Leaf2
<pre> router bgp 65001 router- </pre>	<pre> router bgp 65001 router- id 10.0.2.2 </pre>

Leaf1	Leaf2
<pre> id 10.0.2.1 bestpath as-path multipath-relax address-family ipv4 unicast redistribute connected maximum-paths 2 exit address-family l2vpn evpn advertise- all-vni advertise-pip peer-ip 10.0.2.2 exit </pre>	<pre> bestpath as-path multipath-relax address-family ipv4 unicast redistribute connected maximum-paths 2 exit address-family l2vpn evpn advertise- all-vni advertise-pip peer-ip 10.0.2.1 exit </pre>

10. Configure the BGP spine peer-group.
11. Set timers and enable unnumbered BGP and BFD on the peer-group.
12. Activate both IPv4 unicast and EVPN address families for the spine peer group.
13. Set allowas-in to support the MC-LAG peer's ASN to be received.

Leaf1	Leaf2
<pre> peer-group SPINE advertisement-interval 5 timers 3 9 remote-as external capability extended-nexthop bfd address-family ipv4 unicast activate allowas-in 1 exit address-family l2vpn evpn activate exit exit </pre>	<pre> peer-group SPINE advertisement-interval 5 timers 3 9 remote-as external capability extended-nexthop bfd address-family ipv4 unicast activate allowas-in 1 exit address-family l2vpn evpn activate exit exit </pre>

14. Assign the spine neighbor interfaces to the peer-group.
15. Create the BGP router for each tenant's VRF.
16. Save the configuration.

Leaf1	Leaf2
<pre> neighbor interface Eth 1/53 peer-group SPINE exit neighbor interface Eth 1/56 peer-group SPINE exit exit router bgp 65001 vrf Vrf-tenant1 address-family ipv4 unicast redistribute connected exit address-family l2vpn evpn advertise ipv4 unicast exit exit end write memory </pre>	<pre> neighbor interface Eth 1/53 peer-group SPINE exit neighbor interface Eth 1/56 peer-group SPINE exit exit router bgp 65001 vrf Vrf-tenant1 address-family ipv4 unicast redistribute connected exit address-family l2vpn evpn advertise ipv4 unicast exit exit end write memory </pre>

North-South connectivity - Border leaf configuration

The following diagram highlights the switches that are configured as border leaves. BLeaf25 and BLeaf26 are configured as an MC-LAG pair supporting hosts and provide external connectivity, namely, North-South network traffic connection.

NOTE: This configuration does not include the details for the External Router.

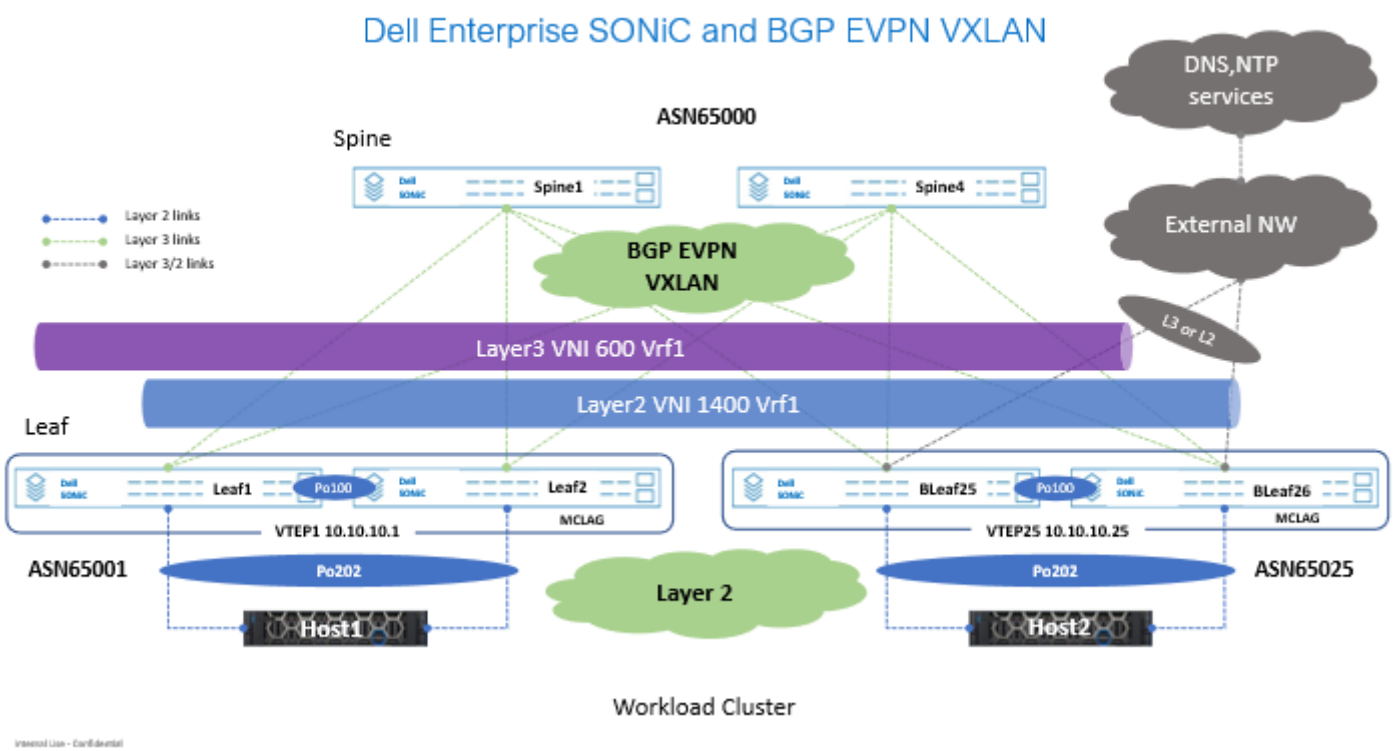


Figure 17. Configuration of border leaves

1. Enter MF-CLI and then configuration mode.
2. Change the interface naming mode to Standard, add a hostname, save the configuration, and exit back to the Linux shell to activate the changes.

BLeaf25	BLeaf26
<pre>sonic-cli configure terminal interface-naming standard hostname BLeaf25 end write memory</pre>	<pre>sonic-cli configure terminal interface-naming standard hostname BLeaf26 end write memory</pre>

3. Re-enter the MF-CLI and configuration mode.
4. Assign a unique router ID on each leaf.
5. Enable IPv6 on spine uplinks to support unnumbered BGP point-to-point links and enable the interfaces.

BLeaf25	BLeaf26
<pre>sonic-cli configure terminal interface loopback 0 description router-id</pre>	<pre>sonic-cli configure terminal interface loopback 0 description router-id</pre>

BLeaf25	BLeaf26
<pre> ip address 10.0.2.25/32 exit interface Eth 1/53 description Spine1 ipv6 enable no shutdown interface Eth 1/56 description Spine4 ipv6 enable no shutdown exit </pre>	<pre> ip address 10.0.2.26/32 exit interface Eth 1/53 description Spine1 ipv6 enable no shutdown interface Eth 1/56 description Spine4 ipv6 enable no shutdown exit </pre>

6. Assign a VRF for each tenant. This POC has only one tenant, but will support multiple tenants with additional VRFs.
7. Configure VLANs 80 and 140 for the hosts, bind to the tenant's VRF, and assign the same anycast-address to each host VLAN.
8. Assign a dedicated VLAN (Vlan60) for a L3 VNI and bind to the tenant's VRF.

 **NOTE:** The VLAN assigned to the L3 VNI does not need an IP address.

BLeaf25	BLeaf26
<pre> ip vrf Vrf-tenant1 interface Vlan 80 ip vrf forwarding Vrf-tenant1 ip anycast-address 192.168.80.254/24 neigh-suppress exit interface Vlan 140 ip vrf forwarding Vrf-tenant1 ip anycast-address 192.168.140.254/24 neigh-suppress exit interface Vlan 60 ip vrf forwarding Vrf-tenant1 exit </pre>	<pre> ip vrf Vrf-tenant1 interface Vlan 80 ip vrf forwarding Vrf-tenant1 ip anycast-address 192.168.80.254/24 neigh-suppress exit interface Vlan 140 ip vrf forwarding Vrf-tenant1 ip anycast-address 192.168.140.254/24 neigh-suppress exit interface Vlan 60 ip vrf forwarding Vrf-tenant1 exit </pre>

9. Configure the MC-LAG peer-link and port channel members.
10. Configure all VLANs on the MC-LAG's peer-link port channel.

BLeaf25	BLeaf26
<pre> interface PortChannel 100 description MCLAG-Peer-Link switchport trunk allowed vlan add 60,80,140 exit interface Eth 1/49 description MCLAG-Peer-Link channel-group 100 no shutdown exit interface Eth 1/51 description MCLAG-Peer-Link channel-group 100 no shutdown exit </pre>	<pre> interface PortChannel 100 description MCLAG-Peer-Link switchport trunk allowed vlan add 60,80,140 exit interface Eth 1/49 description MCLAG-Peer-Link channel-group 100 no shutdown exit interface Eth 1/51 description MCLAG-Peer-Link channel-group 100 no shutdown exit </pre>

CAUTION: When copying and pasting the content from the preceding table, ensure that the underlined monospace text remains on one line. Failure to keep the underlined content on one line results in a line error on the switch.

11. Configure the MC-LAG domain.
12. The loopback IP address that is used for the router-id is also used as the MC-LAG source and peer IP address.
13. Assign the host's port-channel interfaces and assign to the MC-LAG domain.

NOTE: By default, port channels are LACP. Use the `interface PortChannel <num> mode` on command for static LAGs. If changing LAG modes, you must completely remove and reassign the port channel.

NOTE: The spine switches are used for the MC-LAG peer's keep-alive link communication.

BLeaf25	BLeaf26
<pre>mclag domain 1 source-ip 10.0.2.25 peer-ip 10.0.2.26 peer-link PortChannel 100 delay-restore 90 exit interface PortChannel 202 description Host1_PortChannel switchport trunk allowed vlan add 80,140 mclag 1 exit interface Eth 1/3 description Host1_PortChannel channel-group 202 no shutdown exit</pre>	<pre>mclag domain 1 source-ip 10.0.2.26 peer-ip 10.0.2.25 peer-link PortChannel 100 delay-restore 90 exit interface PortChannel 202 description Host1_PortChannel switchport trunk allowed vlan add 80,140 mclag 1 exit interface Eth 1/3 description Host1_PortChannel channel-group 202 no shutdown exit</pre>

14. Enable static anycast-address and assign a unique MAC to be used across all leaf switches.
15. Create a loopback interface for the VTEP IP address. This address must be the same for each MC-LAG peer.
16. Create a VXLAN interface and assign the same VTEP source IP address to each peer.
17. Assign a unique primary-ip to optimize routing for orphan ports and active-standby hosts. The router-id can be used for this purpose.
18. Map the L2 VNI to the host VLANs that are stretched across the fabric.
19. Create a L3 VNI by mapping the dedicated VLAN (Vlan60) to the non-default VRF. The Network Virtualization Overlay (NVO) is automatically assigned to the VXLAN interface.

NOTE: The Network Virtualization Overlay (NVO) is automatically assigned to the VXLAN interface

BLeaf25	BLeaf26
<pre>ip anycast-address enable ip anycast-mac-address 00:00:00:00:01:02 interface Loopback 1 description LogicalVTEP ip address 10.10.10.25/32 exit interface vxlan vtep25 source-ip 10.10.10.25 primary-ip 10.0.2.25 map vni 1400 vlan 140 map vni 600 vlan 60 map vni 600 vrf Vrf-tenant1 exit</pre>	<pre>ip anycast-address enable ip anycast-mac-address 00:00:00:00:01:02 interface Loopback 1 description LogicalVTEP ip address 10.10.10.25/32 exit interface vxlan vtep25 source-ip 10.10.10.25 primary-ip 10.0.2.26 map vni 1400 vlan 140 map vni 600 vlan 60 map vni 600 vrf Vrf-tenant1 exit</pre>

20. Configure the BGP router with ECMP enabled.
21. Configure the underlay's address-family and redistribute the connected networks.
22. Add VNI advertisements to the EVPN address family and assign the MC-LAG peer's primary-ip.

BLeaf25	BLeaf26
<pre> router bgp 65025 router-id 10.0.2.25 bestpath as-path multipath-relax address-family ipv4 unicast redistribute connected maximum-paths 2 exit address-family l2vpn evpn advertise-all-vni advertise-pip peer-ip 10.0.2.26 exit </pre>	<pre> router bgp 65025 router-id 10.0.2.26 bestpath as-path multipath-relax address-family ipv4 unicast redistribute connected maximum-paths 2 exit address-family l2vpn evpn advertise-all-vni advertise-pip peer-ip 10.0.2.25 exit </pre>

23. Configure the BGP spine peer-group.
24. Set timers and enable unnumbered BGP and BFD on the peer-group.
25. Activate both IPv4 unicast and EVPN address families for the spine peer group.
26. Set allowas-in to support MC-LAG peer's ASN to be received.

BLeaf25	BLeaf26
<pre> peer-group SPINE advertisement-interval 5 timers 3 9 remote-as external capability extended-nexthop bfd address-family ipv4 unicast activate allowas-in 1 exit address-family l2vpn evpn activate exit exit </pre>	<pre> peer-group SPINE advertisement-interval 5 timers 3 9 remote-as external capability extended-nexthop bfd address-family ipv4 unicast activate allowas-in 1 exit address-family l2vpn evpn activate exit exit </pre>

27. Assign the spine neighbor interfaces to the peer-group.
28. Create the BGP router for each tenant's VRF.
29. Save the configuration.

BLeaf25	BLeaf26
<pre> neighbor interface Eth 1/53 peer-group SPINE exit neighbor interface Eth 1/56 peer-group SPINE exit exit router bgp 65025 vrf Vrf-tenant1 address-family ipv4 unicast redistribute connected exit address-family l2vpn evpn advertise ipv4 unicast exit exit end </pre>	<pre> neighbor interface Eth 1/53 peer-group SPINE exit neighbor interface Eth 1/56 peer-group SPINE exit exit router bgp 65025 vrf Vrf-tenant1 address-family ipv4 unicast redistribute connected exit address-family l2vpn evpn advertise ipv4 unicast exit exit end </pre>

BLeaf25	BLeaf26
write memory	write memory

Monitoring the POC environment

Enterprise SONiC includes several monitoring capabilities. For this POC, syslog and SNMP are used to monitor the environment. Apply the following configurations to each switch node.

1. If DHCP is not used, configure the out-of-band (OOB) management IP interface and the gateway IP interface for each switch:

```
admin@sonic:~$ sonic-cli
sonic# configure terminal
sonic(config)# interface management 0
sonic(config-if-eth0)# ip address A.B.C.D/len gwaddr A.B.C.D
```

2. Configure the syslog server:

```
logging server A.B.C.D remote-port 514
```

3. Enable the SNMP agent on the switches:

```
sonic(config)# snmp-server agentaddress A.B.C.D
sonic(config)# snmp-server community public
```

4. Configure one or more SNMP server destinations for SNMPv2 traps (UDP/1024):

```
sonic(config)# snmp-server enable trap
sonic(config)# snmp-server host A.B.C.D community public port 1024
```

5. Save the configuration and, optionally, restart the SNMP service:

```
end
write memory
exit

$ sudo systemctl restart snmp
$ systemctl status snmp
```

If you restart the SNMP service, expect 30 seconds for the service to recover.

Verifying the POC

The following table shows the different test cases and the MF-CLI show commands that are used to verify the POC:

Table 2. Proof of Concept features and show commands verification

Feature	Validation step
EBGP Underlay unnumbered	Verify unnumbered BGP neighbors between spine and leaf: <pre>show bgp ipv4 unicast summary show bgp ipv4 unicast neighbors</pre>

Table 2. Proof of Concept features and show commands verification (continued)

Feature	Validation step
	Verify IPv4 loopback advertisement over unnumbered BGP: <pre>show ip route bgp show ip route connected</pre>
	Verify the usage of peer-group in BGP: <pre>show run bgp show bgp peer-group</pre>
	Verify that the ECMP path is established for the advertised loopback interfaces: <pre>show ip route bgp</pre>
EBGP Overlay	Verify that the VXLAN tunnel is formed between the VTEPs: i NOTE: A vxlan tunnel may not form until an active host is attached. <pre>show vxlan tunnel</pre>
	Verify L3 VNIs are getting advertised between the VTEPs <pre>show bgp l2vpn evpn</pre>
	Verify the directly connected IPv4 networks on the VRFs are advertised as well as learned on the associated L3 VNI <pre>show ip route vrf Vrf-tenant1</pre>
	Verify that the IPv4 routes learned on the VRF are redistributed into the overlay network and learn on the peer VTEPs <pre>show ip route vrf Vrf-tenant1</pre>
	Verify the IPv4 data traffic from the connected interface in one VRF in a leaf device to the same VRF across another leaf device: <pre>ping from Host1 to Host2</pre>
	From host1, ping host2 over VLAN 140 and VLAN 50: <pre>Host1\$ ping -I vlan50 192.168.80.102 Host1\$ ping -I vlan140 192.168.80.102 Host1\$ ping -I vlan140 192.168.140.102 Host1\$ ping -I vlan50 192.168.140.102</pre>
	Commands for additional VXLAN information: <pre>show vxlan interface show evpn show evpn vni 1400 show evpn vni 600</pre>
High Availability	Verify that the traffic from the host is switched from one leaf to another leaf during a link failure.
	Verify that the traffic in the leaf switches from one spine to another spine if there is a link failure on the link with active traffic.

Table 2. Proof of Concept features and show commands verification (continued)

Feature	Validation step
	Verify that the traffic in the leaf switches to the peer leaf using the connected link when the link to all spines fails.
SNMP	Verify snmpwalk for the interface MIB.

BGP EVPN VXLAN with Multihoming Configuration

This chapter presents the following topics:

Topics:

- Spine configuration steps
- BGP and BFD unnumbered configuration
- East-West connectivity – Leaf switches configuration
- VRF configuration
- EVPN Multihoming (E-MH) configuration
- North-South connectivity - Border leaf configuration
- Monitoring the PoC environment
- Verifying the PoC

Spine configuration steps

The following diagram highlights the devices configured in this section. Spine1 and Spine4 are configured, demonstrating the use of four spines.

Dell Enterprise SONiC and EVPN Multihoming

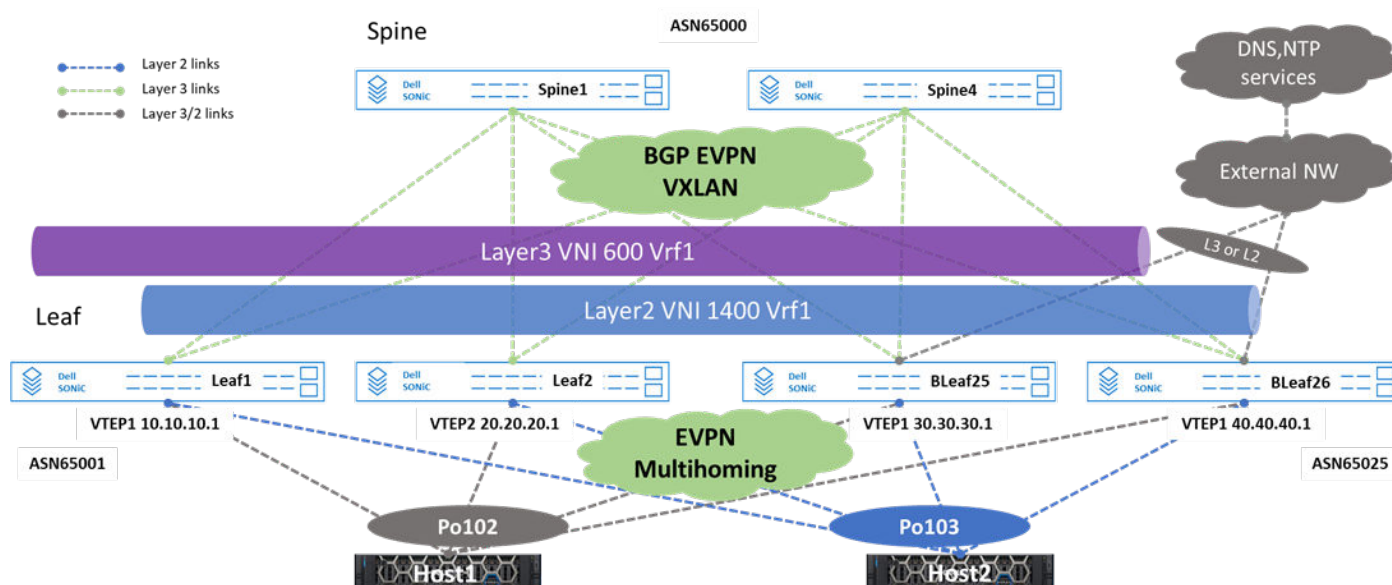


Figure 18. Device configuration

1. Enter MF-CLI and configuration mode.
2. Change interface naming mode to Standard and add a hostname.
3. Exit back to the Linux shell to activate the changes.

Spine1	Spine4
<pre>sonic-cli configure terminal interface-naming standard hostname Spine1 end</pre>	<pre>sonic-cli configure terminal interface-naming standard hostname Spine4 end</pre>

- Re-enter the MF-CLI and configuration mode.
- Assign a loopback interface with a unique router ID for each spine.
- Enable IPv6 on the spine downlinks to support unnumbered BGP point-to-point links and enable the interfaces.

NOTE: Only Spine1 and Spine 4 are configured in this guide. More spines can be added if needed to increase fabric bandwidth.

Spine1	Spine4
<pre>sonic-cli configure terminal interface loopback 0 description router-id ip address 10.0.1.1/32 exit interface Eth 1/1 description Leaf1 ipv6 enable no shutdown interface Eth 1/2 description Leaf2 ipv6 enable no shutdown interface Eth 1/25 description BLeaf25 ipv6 enable no shutdown interface Eth 1/26 description BLeaf26 ipv6 enable no shutdown exit</pre>	<pre>sonic-cli configure terminal interface loopback 0 description router-id ip address 10.0.1.4/32 exit interface Eth 1/1 description Leaf1 ipv6 enable no shutdown interface Eth 1/2 description Leaf2 ipv6 enable no shutdown interface Eth 1/25 description BLeaf25 ipv6 enable no shutdown interface Eth 1/26 description BLeaf26 ipv6 enable no shutdown exit</pre>

BGP and BFD unnumbered configuration

- Configure the BGP router with ECMP enabled.
- Add the IPv4 unicast address-family to advertise the router-ID and paths for two leafs.
- Configure the BGP leaf peer-group. Set timers and enable unnumbered BGP and BFD.
- Activate both IPv4 unicast and EVPN address families for the peer group.
- Assign the leaf neighbor interfaces to the peer-group.
- Save the configuration.

Spine1	Spine4
<pre>router bgp 65000 router-id 10.10.10.1 bestpath as-path multipath-relax address-family ipv4 unicast redistribute connected maximum-paths 2</pre>	<pre>router bgp 65000 router-id 20.20.20.1 bestpath as-path multipath-relax address-family ipv4 unicast redistribute connected maximum-paths 2</pre>

Spine1	Spine4
<pre> exit peer-group LEAF advertisement-interval 5 timers 3 9 remote-as external capability extended-nexthop bfd address-family ipv4 unicast activate exit address-family l2vpn evpn activate exit neighbor interface Eth 1/1 peer-group LEAF exit neighbor interface Eth 1/2 peer-group LEAF exit neighbor interface Eth 1/25 peer-group LEAF exit neighbor interface Eth 1/26 peer-group LEAF end write memory </pre>	<pre> exit peer-group LEAF advertisement-interval 5 timers 3 9 remote-as external capability extended-nexthop bfd address-family ipv4 unicast activate exit address-family l2vpn evpn exit neighbor interface Eth1/1 peer-group LEAF exit neighbor interface Eth1/2 peer-group LEAF exit neighbor interface Eth1/25 peer-group LEAF exit neighbor interface Eth1/26 peer-group LEAF end write memory </pre>

East-West connectivity – Leaf switches configuration

The following diagram highlights the leaf switches that are configured. Leaf1 and Leaf2 are configured as a VTEP with E-MH providing link redundancy to Host1 and Host2.

Dell Enterprise SONiC and EVPN Multihoming

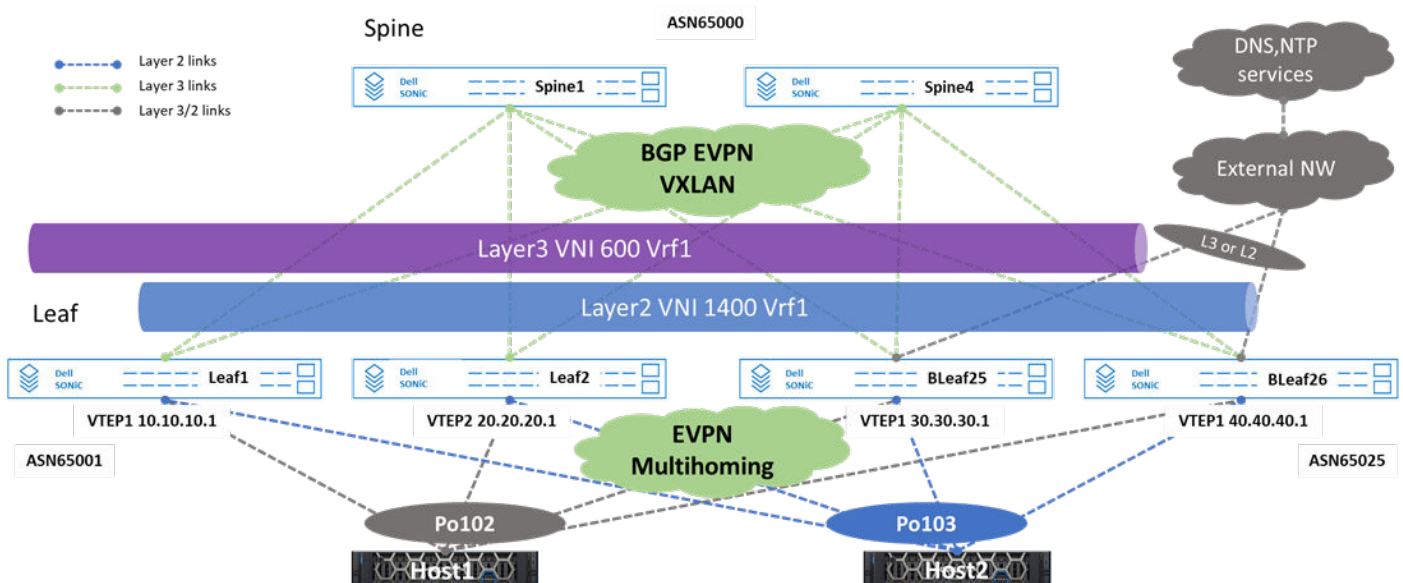


Figure 19. Leaf1 and Leaf2 configuration

1. Enter MF-CLI and then configuration mode.

2. Change the interface naming mode to Standard and add a hostname.
3. Exit back to the Linux shell to activate the changes.

Leaf1	Leaf2
<pre>sonic-cli configure terminal interface-naming standard hostname Leaf1 end</pre>	<pre>sonic-cli configure terminal interface-naming standard hostname Leaf2 end</pre>

4. Re-enter the MF-CLI and configuration mode
5. Assign a loopback interface with a unique router ID on each leaf.
6. Then enable IPv6 on spine uplinks to support unnumbered BGP point-to-point links and enable the interfaces.

Leaf1	Leaf2
<pre>sonic-cli configure terminal interface loopback 0 description router-id ip address 10.10.10.1/32 exit interface Eth 1/53 description Spine1 ipv6 enable no shutdown interface Eth 1/56 description Spine4 ipv6 enable no shutdown exit</pre>	<pre>sonic-cli configure terminal interface loopback 0 description router-id ip address 20.20.20.1/32 exit interface Eth 1/53 description Spine1 ipv6 enable no shutdown interface Eth 1/56 description Spine4 ipv6 enable no shutdown exit</pre>

VRF configuration

1. Configure the vrf.
2. Configure a unique anycast mac address for each leaf pair.
3. Enable `ip anycast` on the leaf pair.
4. Assign a VRF for each tenant. This PoC has only one tenant, but will support multiple tenants with additional VRFs.
5. Configure VLANs 50 and 140 for the hosts, bind to the tenant's VRF and assign the same anycast-address to each host VLAN.
6. Assign a dedicated VLAN (Vlan60) for a L3 VNI and bind to the tenant's VRF.

 **CAUTION:** When copying and pasting the content from the table above, ensure that the underlined monospace text remains on one line. Failure to keep the underlined content on one line results in a line error on the switch.

Leaf1	Leaf2
<pre>ip vrf Vrf-tenant1 ip anycast-mac-address 00:00:00:0a:0b:0c ip anycast-address enable interface Vlan 50 ip vrf forwarding Vrf-tenant1 ip anycast-address 192.168.50.254/24 neigh-suppress exit interface Vlan 140 ip vrf forwarding Vrf-tenant1 ip anycast-address 192.168.140.254/24 neigh-suppress</pre>	<pre>ip vrf Vrf-tenant1 ip anycast-mac-address 00:00:00:0a:0b:0c ip anycast-address-enable ip vrf Vrf-tenant1 interface Vlan 50 ip vrf forwarding Vrf-tenant1 ip anycast-address 192.168.50.254/24 neigh-suppress exit interface Vlan 140 ip vrf forwarding Vrf-tenant1 ip anycast-address 192.168.140.254/24</pre>

Leaf1	Leaf2
<pre> system-mac 00:00:00:11:22:34 evpn ethernet-segment auto-system-mac exit switchport trunk allowed vlan add 50,80,60,140 no shutdown interface Eth 1/1 description Po102 member link channel-group 102 no shutdown exit interface Eth 1/2 description Po103 member link channel-group 103 no shutdown end </pre>	<pre> evpn ethernet-segment auto-system-mac exit switchport trunk allowed vlan add 50,80,60,140 no shutdown interface Eth 1/1 description Po102 member link channel-group 102 no shutdown exit interface Eth 1/2 description Po103 member link channel-group 103 no shutdown end </pre>

BLeaf25	BLeaf26
<pre> sonic-cli configure terminal evpn esi-multihoming startup-delay 300 exit interface PortChannel 102 description Host1-PortChannel system-mac 00:00:00:11:22:33 evpn ethernet-segment auto-system-mac exit switchport trunk allowed vlan add 50,80,60,140 no shutdown exit interface PortChannel 103 description Host2-PortChannel system-mac 00:00:00:11:22:34 evpn ethernet-segment auto-system-mac exit switchport trunk allowed vlan add 50,80,60,140 no shutdown interface Eth 1/1 description Po102 member link channel-group 102 no shutdown exit interface Eth 1/2 description Po103 member link channel-group 103 no shutdown end </pre>	<pre> sonic-cli configure terminal evpn esi-multihoming startup-delay 300 exit interface PortChannel 102 description Host1-PortChannel system-mac 00:00:00:11:22:33 evpn ethernet-segment auto-system-mac exit switchport trunk allowed vlan add 50,80,60,140 no shutdown exit interface PortChannel 103 description Host2-PortChannel system-mac 00:00:00:11:22:34 evpn ethernet-segment auto-system-mac exit switchport trunk allowed vlan add 50,80,60,140 no shutdown interface Eth 1/1 description Po102 member link channel-group 102 no shutdown exit interface Eth 1/2 description Po103 member link channel-group 103 no shutdown end </pre>

3. Create a loopback interface for the VTEP IP address.
4. Create a VXLAN interface and assign the same loopback interface as the VTEP source IP address to each peer. Notice, each VTEP is configured with a different IP.
5. Assign a unique primary-ip to optimize routing for orphan ports and active-standby hosts. The router-id can be used for this purpose.
6. Map the L2 VNI to the host VLANs that are stretched across the fabric.
7. Create a L3 VNI by mapping the dedicated VLAN (Vlan60) to the non-default VRF.

Leaf1	Leaf2
<pre> interface vxlan vtep1 source-ip 10.10.10.1 map vni 1400 vlan 140 map vni 600 vlan 60 map vni 600 vrf Vrf-tenant1 exit </pre>	<pre> interface vxlan vtep1 source-ip 20.20.20.1 map vni 1400 vlan 140 map vni 600 vlan 60 map vni 600 vrf Vrf-tenant1 exit </pre>

 **NOTE:** The Network Virtualization Overlay (NVO) is automatically assigned to the VXLAN interface.

8. Configure the BGP router with ECMP enabled.
9. Configure the underlay's address-family and redistribute the connected networks.
10. Add VNI advertisements to the EVPN address family and assign the loopback ip as the primary IP address

Leaf1	Leaf2
<pre> router bgp 65001 router-id 10.0.2.1 bestpath as-path multipath-relax address-family ipv4 unicast redistribute connected maximum-paths 2 exit address-family l2vpn evpn advertise-all-vni advertise-pip peer-ip 10.0.2.2 exit </pre>	<pre> router bgp 65001 router-id 10.0.2.2 bestpath as-path multipath-relax address-family ipv4 unicast redistribute connected maximum-paths 2 exit address-family l2vpn evpn advertise-all-vni advertise-pip peer-ip 10.0.2.1 exit </pre>

11. Configure the BGP spine peer-group.
12. Set timers and enable unnumbered BGP and BFD on the peer-group.
13. Activate both IPv4 unicast and EVPN address families for the spine peer group.
14. Set allowas-in.

Leaf1	Leaf2
<pre> peer-group SPINE advertisement-interval 5 timers 3 9 remote-as external capability extended-nexthop bfd address-family ipv4 unicast activate allowas-in 1 exit address-family l2vpn evpn activate exit </pre>	<pre> peer-group SPINE advertisement-interval 5 timers 3 9 remote-as external capability extended-nexthop bfd address-family ipv4 unicast activate allowas-in 1 exit address-family l2vpn evpn activate exit </pre>

15. Assign the spine neighbor interfaces to the peer-group.
16. Create the BGP router for each tenant's VRF.
17. Save the configuration.

Leaf1	Leaf2
<pre> neighbor interface Eth 1/53 peer-group SPINE exit </pre>	<pre> neighbor interface Eth 1/53 peer-group SPINE exit </pre>

Leaf1	Leaf2
<pre> neighbor interface Eth 1/56 peer-group SPINE exit router bgp 65001 vrf Vrf-tenant1 address-family ipv4 unicast redistribute connected exit address-family l2vpn evpn advertise ipv4 unicast end write memory </pre>	<pre> neighbor interface Eth 1/56 peer-group SPINE exit router bgp 65001 vrf Vrf-tenant1 address-family ipv4 unicast redistribute connected exit address-family l2vpn evpn advertise ipv4 unicast end write memory </pre>

North-South connectivity - Border leaf configuration

The following diagram highlights the switches that are configured as border leaves. BLeaf25 and BLeaf26 are configured as E-MH switches connected to Host1 and Host2 over the different port-channels and to an external router/switch providing North-South connectivity.

NOTE: This configuration does not include the details for the External Router.

Dell Enterprise SONiC and EVPN Multihoming

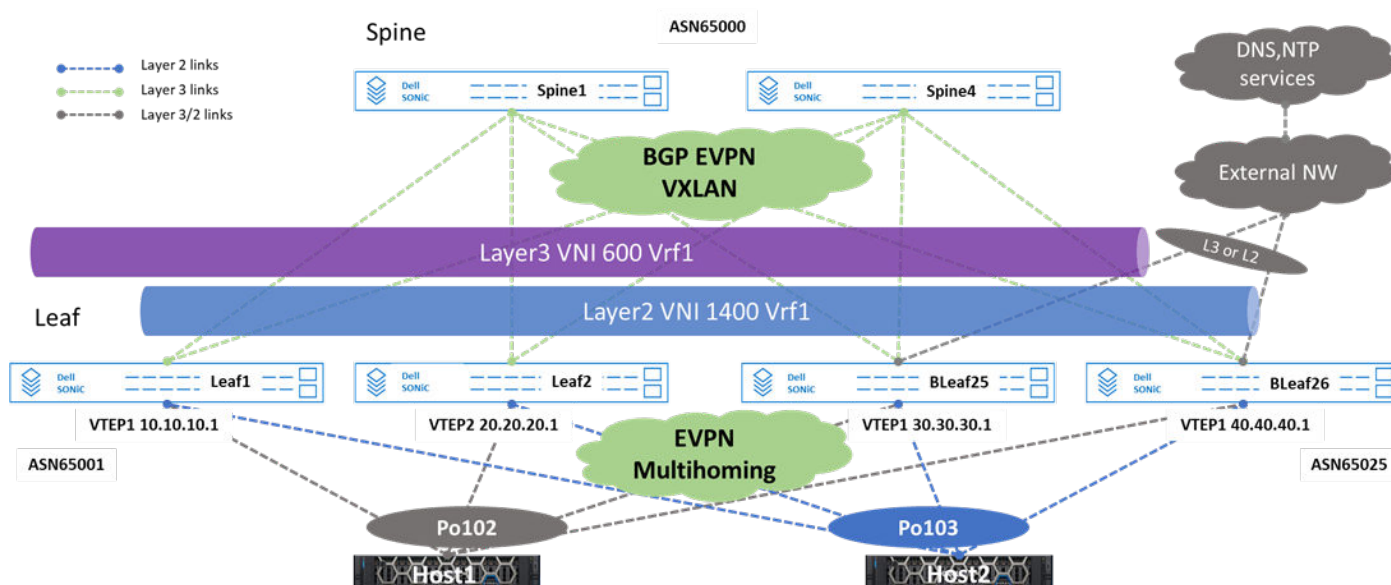


Figure 21. Configuration of border leaves

1. Enter MF-CLI and then configuration mode.
2. Change the interface naming mode to Standard, add a hostname, save the configuration, and exit back to Linux shell to activate the changes.

BLeaf25	BLeaf26
<pre> sonic-cli configure terminal interface-naming standard </pre>	<pre> sonic-cli configure terminal interface-naming standard </pre>

BLeaf25	BLeaf26
<pre>hostname BLeaf25 end write memory</pre>	<pre>hostname BLeaf26 end write memory</pre>

3. Re-enter the MF-CLI and configuration mode.
4. Assign a unique router ID on each leaf.
5. Then enable IPv6 on spine uplinks to support unnumbered BGP point-to-point links and enable the interfaces.

BLeaf25	BLeaf26
<pre>sonic-cli configure terminal interface loopback 0 description router-id ip address 10.0.2.25/32 exit interface Eth 1/53 description Spine1 ipv6 enable no shutdown interface Eth 1/56 description Spine4 ipv6 enable no shutdown exit</pre>	<pre>sonic-cli configure terminal interface loopback 0 description router-id ip address 10.0.2.26/32 exit interface Eth 1/53 description Spine1 ipv6 enable no shutdown interface Eth 1/56 description Spine4 ipv6 enable no shutdown exit</pre>

6. Assign a VRF for each tenant. This PoC only has one tenant but will support multiple tenants with additional VRFs.
7. Configure VLANs 80 and 140 for the hosts, bind to the tenant's VRF and assign the same anycast-address to each host VLAN.
8. Assign a dedicated VLAN (Vlan60) for a L3 VNI and bind to the tenant's VRF.



NOTE: The VLAN assigned to the L3 VNI does not need an IP address.

BLeaf25	BLeaf26
<pre>ip vrf Vrf-tenant1 ip anycast-mac-address 00:00:00:00:01:02 ip anycast-address enable ip vrf Vrf-tenant1 interface Vlan 80 ip vrf forwarding Vrf-tenant1 ip anycast-address 192.168.80.254/24 neigh-suppress exit interface Vlan 140 ip vrf forwarding Vrf-tenant1 ip anycast-address 192.168.140.254/24 neigh-suppress exit interface Vlan 60 ip vrf forwarding Vrf-tenant1 exit</pre>	<pre>ip vrf Vrf-tenant1 ip anycast-mac-address 00:00:00:00:01:02 ip anycast-address enable interface Vlan 80 ip vrf forwarding Vrf-tenant1 ip anycast-address 192.168.80.254/24 neigh-suppress exit interface Vlan 140 ip vrf forwarding Vrf-tenant1 ip anycast-address 192.168.140.254/24 neigh-suppress exit interface Vlan 60 ip vrf forwarding Vrf-tenant1 exit</pre>



CAUTION: When copying and pasting the content from the table above, ensure that the underlined monospace text remains on one line. Failure to keep the underlined content on one line results in a line error on the switch.

9. Enable static anycast-address and assign a unique MAC to be used across all leaf switches.
10. Create a loopback interface for the VTEP IP address.
11. Create a VXLAN interface and assign the loopback interface IP as the VTEP source IP address to each peer.
12. Assign a unique primary-ip to optimize routing for orphan ports and active-standby hosts. The router-id can be used for this purpose.
13. Map the L2 VNI to the host VLANs that are stretched across the fabric.
14. Create a L3 VNI by mapping the dedicated VLAN (Vlan60) to the non-default VRF. The Network Virtualization Overlay (NVO) is automatically assigned to the VXLAN interface.

NOTE: The Network Virtualization Overlay (NVO) is automatically assigned to the VXLAN interface

BLeaf25	BLeaf26
<pre> interface Loopback 1 description LogicalVTEP ip address 30.30.30.1/32 exit interface vxlan vtep25 source-ip 30.30.30.1 primary-ip 30.30.30.1 map vni 1400 vlan 140 map vni 600 vlan 60 map vni 600 vrf Vrf-tenant1 exit </pre>	<pre> interface Loopback 1 description LogicalVTEP ip address 40.40.40.1/32 exit interface vxlan vtep25 source-ip 40.40.40.1 primary-ip 40.40.40.1 map vni 1400 vlan 140 map vni 600 vlan 60 map vni 600 vrf Vrf-tenant1 exit </pre>

15. Configure the BGP router with ECMP enabled.
16. Configure the underlay's address-family and redistribute the connected networks.
17. Add VNI advertisements to the EVPN address family and use the loopback as the source IP address.

BLeaf25	BLeaf26
<pre> router bgp 65025 router-id 10.0.2.25 bestpath as-path multipath-relax address-family ipv4 unicast redistribute connected maximum-paths 2 exit address-family l2vpn evpn advertise-all-vni advertise-pip peer-ip 10.0.2.26 exit </pre>	<pre> router bgp 65025 router-id 10.0.2.26 bestpath as-path multipath-relax address-family ipv4 unicast redistribute connected maximum-paths 2 exit address-family l2vpn evpn advertise-all-vni advertise-pip peer-ip 10.0.2.25 exit </pre>

18. Configure the BGP spine peer-group.
19. Set timers and enable unnumbered BGP and BFD on the peer-group.
20. Activate both IPv4 unicast and EVPN address families for the spine peer group.
21. Set allowas-in.

BLeaf25	BLeaf26
<pre> peer-group SPINE advertisement-interval 5 timers 3 9 remote-as external capability extended-nexthop bfd address-family ipv4 unicast activate allowas-in 1 </pre>	<pre> peer-group SPINE advertisement-interval 5 timers 3 9 remote-as external capability extended-nexthop bfd address-family ipv4 unicast activate allowas-in 1 </pre>

BLeaf25	BLeaf26
<pre>exit address-family l2vpn evpn activate exit</pre>	<pre>exit address-family l2vpn evpn activate exit</pre>

22. Assign the spine neighbor interfaces to the peer-group.
23. Create the BGP router for each tenant's VRF.
24. Save the configuration.

Monitoring the PoC environment

Enterprise SONiC includes several monitoring capabilities. For this PoC, syslog and SNMP are used to monitor the environment. Apply the following configurations to each switch node:

1. If DHCP is not used, configure the out-of-band (OOB) management IP interface and the gateway IP interface for each switch:

```
admin@sonic:~$ sonic-cli sonic# configure terminal
sonic(config)# interface management 0
sonic(config-if-eth0)# ip address A.B.C.D/len gwaddr A.B.C.D
```

2. Configure the syslog server:

```
logging server A.B.C.D remote-port 514
```

3. Enable the SNMP agent on the switches:

```
sonic(config)# snmp-server agentaddress A.B.C.D sonic(config)#
snmp-server community public
```

4. Configure one or more SNMP server destinations for SNMPv2 traps (UDP/1024):

```
sonic(config)# snmp-server enable trap
sonic(config)# snmp-server host A.B.C.D community public port 1024
```

5. Save the configuration and, optionally, restart the SNMP service:

```
end write
memory exit

$ sudo systemctl restart snmp
$ systemctl status snmp
```

If you restart the SNMP service, allow 30 seconds for the service to recover.

Verifying the PoC

Verify that the PoC procedure is successful.

The following table shows the different test cases and MF-CLI show commands that the Dell team used to verify the PoC:

Table 3. Proof of Concept features and show commands verification

Feature	Validation steps
EBGP Underlay unnumbered	Verify unnumbered BGP neighbors between spine and leaf: <pre>show bgp ipv4 unicast summary show bgp ipv4 unicast neighbors</pre>
	Verify IPv4 loopback advertisement over unnumbered BGP: <pre>show ip route bgp show ip route connected</pre>
	Verify the usage of peer-group in BGP: <pre>show run bgp show bgp peer-group</pre>
	Verify that the ECMP path is established for the advertised loopback interfaces: <pre>show ip route bgp</pre>
EBGP Overlay	Verify that the VXLAN tunnel is formed between the VTEPs. i NOTE: A VXLAN tunnel may not form until an active host is attached. <pre>show vxlan tunnel</pre>
	Verify that L3 VNIs are getting advertised between the VTEPs: <pre>show bgp l2vpn evpn</pre>
	Verify the directly connected IPv4 networks on the VRFs are advertised as well as learned on the associated L3 VNI: <pre>show ip route vrf Vrf-tenant1</pre>
	Verify that the IPv4 routes learned on the VRF are redistributed into the overlay network and learn on the peer VTEPs: <pre>show ip route vrf Vrf-tenant1</pre>
	Verify the IPv4 data traffic from the connected interface in one VRF in a leaf device to the same VRF across another leaf device: <pre>ping from Host1 to Host2</pre>
	From host1, ping host2 over VLAN 140 and VLAN 50: <pre>Host1\$ ping -I vlan50 192.168.80.102 Host1\$ ping -I vlan140 192.168.80.102 Host1\$ ping -I vlan140 192.168.140.102 Host1\$ ping -I vlan50 192.168.140.102</pre>

Table 3. Proof of Concept features and show commands verification (continued)

Feature	Validation steps
	Commands for additional VXLAN information: <pre>show vxlan interface show evpn show evpn vni 1400 show evpn vni 600</pre>
High Availability	Verify that the traffic from the host is switched from one leaf to another leaf during a link failure.
	Verify that the traffic in the leaf switches from one spine to another spine if there is a link failure on the link with active traffic.
	Verify that the traffic in the leaf switches to the peer leaf using the connected link when the link to all spines fails.
SNMP	Verify <code>snmpwalk</code> for the interface MIB.
EVPN multihoming status	Verify EVPN multihoming status <pre>show evpn es detail show bgp l2vpn evpn</pre>
	Verify any EVPN inconsistencies <pre>show bgp l2vpn evpn es-evi detail show bgp l2vpn es detail</pre>

Host Configuration

This deployment guide assumes ESXi servers with VMs attached to the leaf switches. This appendix provides alternative configurations for bare metal servers using the Ubuntu operating system.

Topics:

- [Ubuntu host configuration](#)

Ubuntu host configuration

The following code snippet is an example of Ubuntu hosts with LACP and tagged VLAN interfaces.

Edit the `/etc/netplan/50-cloud-init.yaml` file.

```
$ sudo vi /etc/netplan/50-cloud-init.yaml

network:
  version: 2
  renderer: network

  ethernets:
    eports:
      match:
        name: ens*

  bonds:
    bond0:
      interfaces: [eports]
      parameters:
        mode: 802.3ad
        lacp-rate: slow
        mii-monitor-interval: 100

  vlans:
    bond0.140:
      id: 140
      link: bond0
      dhcp4: no
      addresses: [192.168.140.102/24]
      gateway4: 192.168.140.254

    bond0.50:
      id: 50
      link: bond0
      dhcp4: no
      addresses: [192.168.50.102/24]
      gateway4: 192.168.50.254

:wq!
```

Follow with:

```
sudo netplan apply-d
sudo reboot
```

POC End Host Configurations

This appendix lists the configurations of Dell switches as end hosts to demonstrate end-host connectivity into the POC environment and to show the connection between the end hosts. For successful validation, the end hosts must be active.

This appendix presents the following topics:

Topics:

- [Dell switch \(Host1\)](#)
- [Dell switch \(Host2\)](#)

Dell switch (Host1)

If servers are not available, hosts may be emulated with OS10 switches. The following configurations are available for reference:

- Host1
- Host2

```
config t
hostname Host1
spanning-tree disable

interface vlan 50
ip address 192.168.50.101/24
exit

interface vlan 140
ip address 192.168.140.101/24
exit

interface portchannel 1
switchport mode trunk
switchport trunk allowed vlan 50,140
no shut
exit

interface ethernet 1/1/1
no switchport
channelgroup 1 mode auto
no shut
exit

interface ethernet 1/1/2
no switchport
channelgroup 1 mode auto
no shut
exit

ip route 0.0.0.0/0 192.168.50.254

end
write memory
```

Dell switch (Host2)

```
config t
hostname Host2
spanning-tree disable

interface vlan 80
ip address 192.168.50.102/24
exit

interface vlan 140
ip address 192.168.140.102/24
exit

interface portchannel 1
switchport mode trunk
switchport trunk allowed vlan 80,140
no shut
exit

interface ethernet 1/1/1
no switchport
channelgroup 1 mode auto
no shut
exit

interface ethernet 1/1/2
no switchport
channelgroup 1 mode auto
no shut
exit

ip route 0.0.0.0/0 192.168.80.254

end
write memory
```

Additional Information

Topics:

- [Dell Technologies documentation](#)
- [Support and feedback](#)

Dell Technologies documentation

The following Dell Technologies documentation provides additional relevant information. Access to these documents depends on your login credentials. If you do not have access to a document, contact your Dell Technologies representative.

- [Networking Solutions Info Hub](#)
- [Enterprise SONiC Documents](#) (Employee or Partner account login credentials are required)
- [Dell Demo Center Networking Hands on Labs](#)

Support and feedback

For technical support, go to [Dell Support](#) or call (USA) 1-800-945-3355.

To provide feedback on the quality and usefulness of this publication, write to Dell_Networking_Solutions@Dell.com.